

Abhiram Tilak

Undergraduate, Computer Science Major, Minor in Natural Sciences, IIIT Hyderabad

abhiram@research.iiit.ac.in | GitHub | Student Life Page

Profile

Junior year at IIIT-H, Researcher at CCNSB.
Specialized in Machine Learning for High-Energy Physics and Particle Physics.
Research interests include High-Energy Physics Phenomenology, Machine Learning in Natural Sciences, and Quantum Information Theory.

Education

IIIT Hyderabad 2022 - Present
Undergraduate Student, Computer Science & Natural Sciences

Publications

Conference Submissions
HEP-JEPA: A foundation model for collider physics using Joint Embedding Predictive Architecture. (Under review, submitted to ICML 2025)

Work Experience

IT Office, IIIT Hyderabad 2024 - 2025
Institute Student Web Administrator

Women In Silicon Hardware (WISH), Talentsprint October - November 2024
Mentor for course Introduction to linux and scripting tools.

Libvirt Summer 2024
Google Summer of Code 2024 Contributor

Projects:

Campus Apps and Portals

My-IIITH App: Android app for campus services (attendance, transcripts, notifications, timetable, mess registrations, etc.). I am one of the 4 original maintainers of this project, and is currently used by a large proportion of students in campus.

Life Portal/Website: Core maintainer of student management web portal for budget approvals, room booking, and student body management.

Coursework

Operating Systems and Networks 2024
BISH: Custom shell in C with standard shell functionalities.
Enhanced-xv6: Enhancements and algorithm implementations for MIT's xv6.
NFS: Custom Network Storage Interface (group project, 4 contributors).

Advanced Computer Architecture
Scoreboarding & Register Renaming: Forked Flute to implement Scoreboarding and Register Renaming.

Data and Application 2023
City Management Database System: Database system for city management (5-member project).

Other Projects

Hackathons and events 2023
Psyche-Screener: AI-based interview screening tool (2nd place, Megathon).

Personal Projects
Comfyline Prompt: Customizable ZSH prompt (listed in Oh My Zsh Wiki).

Roles & Leadership

System Administrator, **SLC-Tech Team** 2024 - Present
News & Content Writer, **Hacking Club (1337x)** 2023 - 2024
Coordinator, **Open Source Development Group (OSDG)** 2022 - 2024

Last updated: January 2025